

KLEBSIELLA PNEUMONIAE CARBAPENEM RESISTANCE PREDICTION

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Abstract

One of the most significant health crisis globally is the emergence of antimicrobial resistance which is especially a big problem in hospital acquired infections to *Klebsiella pneumoniae*. Carbapenem-resistant *K. pneumoniae* (CRKP) strains are particularly threatening as carbapenems are regarded as "last resort" antibiotics. Antibiotic susceptibility testing is a standard lab test that takes 24 to 72 hours to complete, which can make the difference between life and death in cases of life-threatening infections like sepsis.

In this work, the authors proposed a machine learning driven computational approach to predict the carbapenem resistance directly from the bacterial DNA sequences. The proposed system is a pipeline that fetches the β -lactamase gene sequences from the NCBI nucleotide database, does rigorous preprocessing, leakage auditing, homology reduction, 4-mer feature extraction using TF-IDF, and classifies using multiple machine learning algorithms such as Random Forest, XGBoost, Logistic Regression, HistGradientBoosting, and LinearSVC.

One of the key challenges in this research is to avoid data leakage and over fitting due to highly similar gene variants. Hence, a smart pipeline with aggressive cosine similarity filtering and strict regularization is added to the pipeline. The results obtained in the experiments show a realistic and very high level of prediction, with the accuracy of the ROC-AUC curve at 0.995 and the F1-score at 0.95 for the model built with the Random Forest algorithm. The proposed framework shows that the patterns of the genome sequence are adequate to efficiently predict the carbapenem resistance and could serve as a part of preliminary screening systems in clinical bioinformatics applications.

1. Introduction

AMR is an emerging health-care threat globally. *Klebsiella pneumoniae* is one of the most clinically relevant resistant pathogens that cause blood stream infections, pneumonia, urinary tract infections and sepsis in hospitalized patients.

The carbapenems are a class of antibiotics which are last-resort drugs for treating multidrug-resistant bacterial infections. However, treatment options have been severely limited and mortality rates high with the emergence of carbapenem resistant *K. pneumoniae* (CRKP) strains. Carbapenemase-producing genes which mediate resistance include:

- blaKPC
- blaNDM
- blaOXA-48
- blaVIM
- blaIMP

Traditional methods for antimicrobial susceptibility testing involve biochemical testing and culturing in a laboratory and can take days to complete. In severe infections like septic shock, the decision to treat needs to be made quickly, or death may result.

The recent developments of bioinformatics and machine learning algorithms allow for the genomic sequence analysis of fast and accurate resistance prediction. DNA sequences can be analysed *in silico* to detect resistance associated genomic signatures, without waiting for the results of laboratory culture.

This project aims to create leakage controlled machine learning pipeline for predicting bacterial resistance to carbapenems based on the DNA k-mer features encoded using TF-IDF on beta-lactamase gene sequences, in a realistic manner.

2. Literature Survey

With the growing resistance of bacterial pathogens to antibiotics, antimicrobial resistance (AMR) has become a large scale health problem around the globe. Prestinaci et al. spoke about the worldwide effects of AMR and the necessity of better diagnostic and prediction systems for resistant bacterial infections [1]. *Klebsiella pneumoniae* is one of the most clinically relevant hospital associated pathogens due to its close relationship with carbapenem resistance and mechanisms of multidrug resistance. In AMR research, the genomic complexities and the accessory genome of *Klebsiella pneumoniae* were pointed out by Martin and Bachman [2] for their importance in this study.

With recent advances in machine learning and genomics sequencing, computational methods can now be developed for directly predicting antimicrobial resistance from DNA sequences. Recently, Arango-Argoty et al. have proposed deep learning framework DeepARG to predict antibiotic resistance genes from metagenomic sequence data [3]. Their study showed the feasibility of using AI to find genomic patterns associated with resistance. Likewise, Su et al. introduced genome-based prediction methods for antibiotic resistance of bacteria, based on the genomic sequence and machine learning approaches [4]. They found that their study on genomic feature learning can facilitate computational AMR prediction quickly.

DNA sequencing features have also been used to aid machine learning in resistance classification. In order to predict TB drug resistance, Yang et al. have designed machine learning models using DNA sequence data resulting in the demonstration of the effectiveness of DNA based classification methods [5]. Nguyen et al. also investigated the use of genomic machine learning methods to accurately predict antimicrobial minimum inhibitory concentrations (MICs), based on genomic features derived from bacterial sequences [6]. The advantage of using ML models to learn resistance-associated genomic patterns from sequence-based representations was emphasized in these studies.

The application of whole genome sequence analysis for computational phenotyping and identification of resistance biomarkers has grown in significance. The authors of [7] proposed methods to perform predictive computational phenotyping based on genome comparisons and

showed successful predictions of resistance with machine learning classification techniques. Boolchandani et al. reviewed the different antimicrobial resistance detection methods based on sequencing and the increasing role of genomic sequencing technologies in antimicrobial resistance surveillance and resistance analysis [8].

Deep learning has been the source of great promise in bioinformatics and genomic sequence analysis, too. Li et al. summarized the different applications of deep learning in biological sequence learning and pointed out future potential applications of deep learning in genomic prediction systems [9]. Moreover, Nguyen et al. introduced an in silico minimal inhibitory concentration (MIC) prediction model for *Klebsiella pneumoniae* based on genomic data analysis [10]. They showed that their system could effectively be used for computational resistance prediction of clinically relevant pathogens.

Overall, the literature reviewed collectively shows that the extraction of genomic features and machine learning methods are suitable tools for helping predict antimicrobial resistance. Inspired by the above-mentioned studies, the work presented in this thesis is concerned with the prediction of carbapenem resistance in *Klebsiella pneumoniae* by using 4-mer genomic feature extraction using TF-IDF and comparative machine learning classification.

3. Materials and Methods

3.1 Dataset Collection

Sequences of the beta-lactamase gene were retrieved from the nucleotide database at the National Center for Biotechnology Information (NCBI) using the Biopython Entrez API. Both carbapenem-resistant and non-carbapenem beta-lactamase genes were the target of queries.

Resistant Gene Categories

- blaKPC
- blaNDM
- blaOXA-48
- blaVIM
- blaIMP

Non-Resistant Gene Categories

- blaTEM
- blaSHV
- blaCTX-M
- blaOXA-1

The sequences were downloaded in FASTA format in batches of 50 to avoid NCBI API rate limitations.

3.2 Workflow of the Proposed System

The complete workflow consists of the following stages:

Block Diagram of the Proposed System

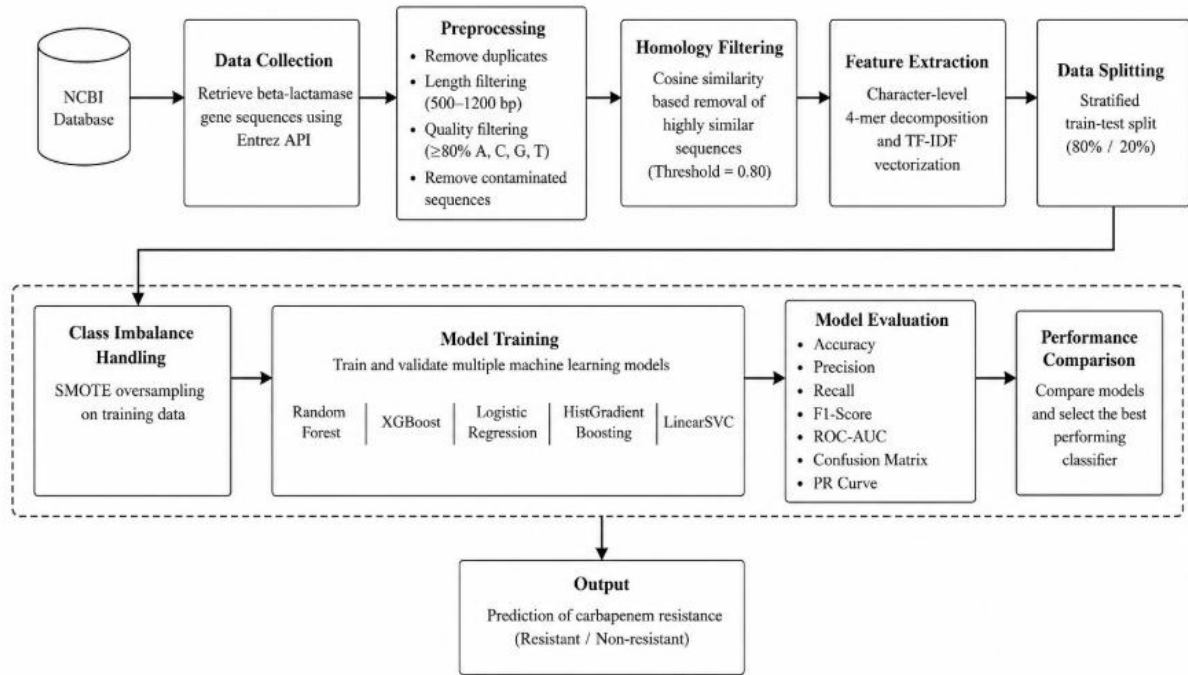


Figure 1: System Workflow

3.3 Preprocessing

Following several pre-processing operations were performed:

a. Duplicate Removal

- Duplicate DNA sequences were filtered out using pandas duplicate filtering to get exact matches.

b. Length Filtering

- Only sequences of 500–1200 bp were kept to avoid fragmented and noisy assemblies.

c. Quality Filtering

- Sequences with less than 80% valid nucleotides (A, C, G, T) were filtered.

d. Contamination Removal

- To prevent label contamination, sequences that contained misleading keywords related to carbapenems in the susceptible class were removed from the sequences.

3.4 Leakage Reduction and Homology Filtering

This project's major contribution is in terms of leakage prevention.

Many publicly available genomic resources have almost identical variants of the same resistance genes. In the case of similar variants in both training data and test data, the model learns "patterns" rather than learning generalizable biology. To prevent this:

- TF-IDF vectors at Character level were constructed.
- The sequences were compared using cosine similarity.
- Sequences with similarity > 0.80 were aggressively filtered.

This is the same as the homology reduction done in CD-HIT, and it greatly enhances the realism of evaluation.

3.5 Feature Extraction

This representation shows biologically meaningful sequence motifs in low dimensional form.

Example:

DNA Sequence:

ATCGTAGC

4-mers:

- ATCG
- TCGT
- CGTA
- GTAG
- TAGC

The TF-IDF vectorizer used:

- analyzer = "char"
- ngram_range = (4,4)
- max_features = 3000
- sublinear_tf = True
- min_df = 3

3.6 Machine Learning Models

Model	Purpose
Random Forest	Ensemble tree classifier
XGBoost	Gradient boosting model
Logistic Regression	Linear baseline classifier
HistGradientBoosting	Efficient boosting classifier
LinearSVC	Margin-based classifier

The parameters for regularization were carefully tuned to prevent overfitting.

3.7 Model Architecture

Proposed Architecture

1. NCBI Sequence Retrieval
2. Sequence Cleaning and Validation
3. Leakage Reduction via Cosine Similarity
4. TF-IDF 4-mer Encoding
5. SMOTE Balancing
6. ML Classification
7. Evaluation Metrics Generation

4. Results and Discussion

4.1 Dataset Distribution

After preprocessing and homology filtering:

- Resistant sequences: 100
- Non-carbapenem sequences: 100
- Final balanced dataset: 200 sequences

There was no change in the class distribution following the filtering process.

4.2 Leakage Audit Analysis

The following leakage analysis was carried out with:

- Length distribution
- GC content distribution
- 1-mer base frequency ROC-AUC

Results showed:

- No significant length loss
- No significant GC bias,
- No leakage of extreme sequences in composition.

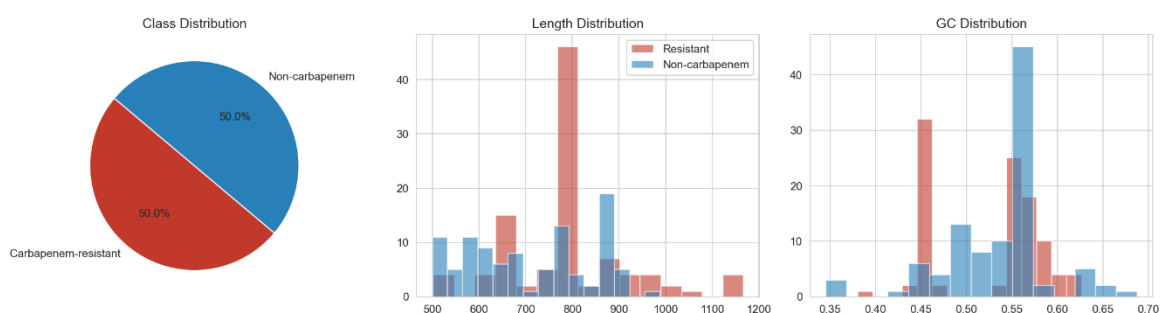


Figure 2: Leakage audit

This validates that the pipeline extracted meaningful features from the dataset, and not dataset-specific features.

4.3 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Avg-Precision	CV-F1 (mean)	CV-F1 (std)
Random Forest	0.95	1	0.9	0.9474	0.995	0.9955	0.9743	0.0167
XGBoost	0.925	1	0.85	0.9189	0.9725	0.9823	0.9743	0.023
Logistic Regression	0.8	0.75	0.9	0.8182	0.965	0.9736	0.9106	0.0245
HistGradientBoosting	0.9	0.9444	0.85	0.8947	0.9575	0.9709	0.9795	0.0196
LinearSVC	0.875	0.8571	0.9	0.878	0.965	0.9737	0.9309	0.0285

Figure 3: Model Performance Comparison

Random Forest classifier gave the best overall performance.

4.4 ROC Curve Analysis

All models demonstrated good discriminatory power in ROC curves.

- The maximum ROC-AUC value obtained was 0.995 by Random Forest.
- XGBoost achieved ROC-AUC = 0.972.
- The Logistic Regression algorithm gave the ROC-AUC = 0.965.

The vectors with high ROC-AUC values represents good separation between resistant and non-resistant sequences.

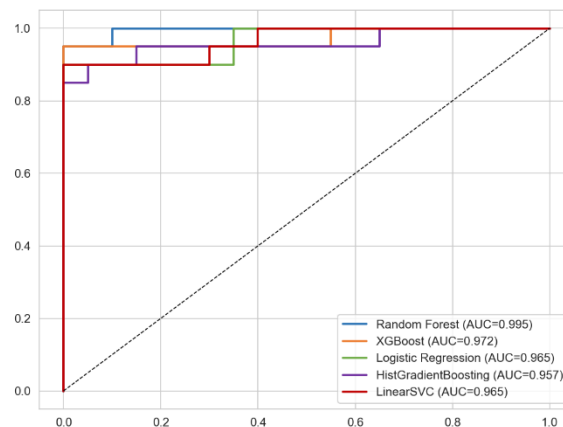


Figure 4: ROC-AUC Curves

4.5 Precision-Recall Curve Analysis

The clinical impact of false-negative predictions is particularly relevant to antimicrobial resistance prediction, particularly for precision-recall analysis.

Random Forest showed the best results in terms of Average Precision (AP = 0.995), demonstrating good resistant-sequence detection capacity.

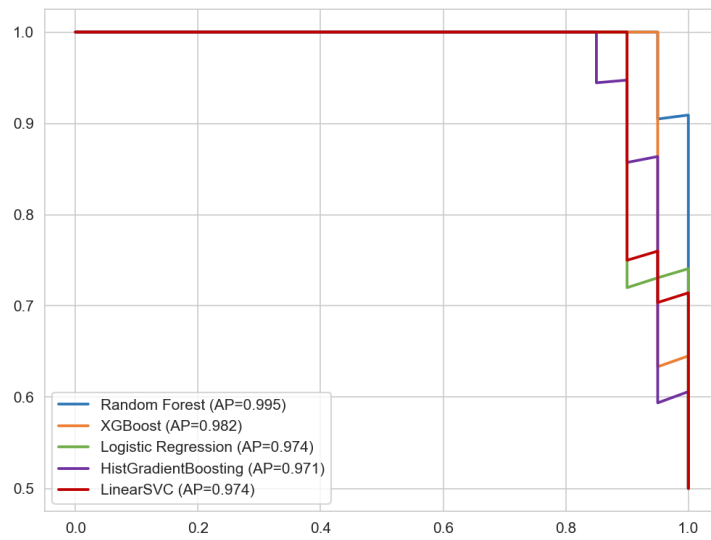


Figure 5: PR Curve Analysis

4.6 Confusion Matrix Analysis

Random Forest

- Correct resistant predictions: 18
- Correct non-resistant predictions: 20
- False positives: 0
- False negatives: 2

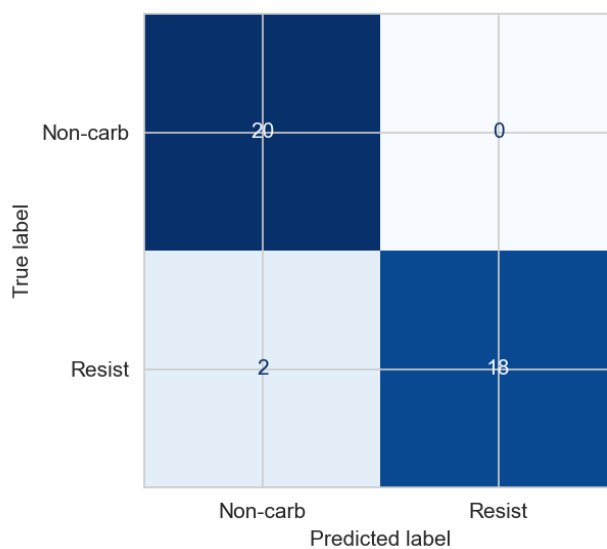


Figure 6: Random Forest Confusion Matrix

XGBoost

- Very high precision
- Slightly lower recall than Random Forest

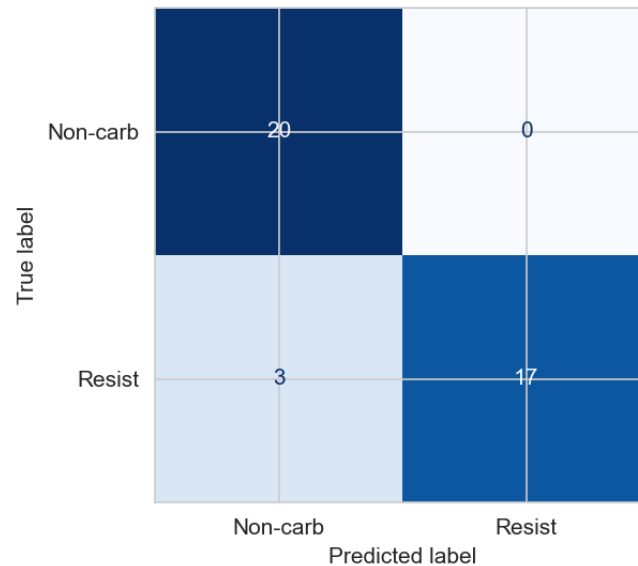


Figure 7: XGBoost Confusion Matrix

Logistic Regression

- Higher false positives due to linear decision boundary limitations

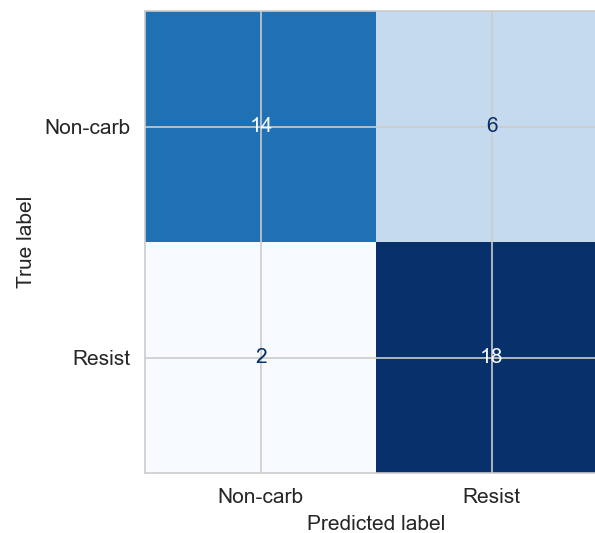


Figure 8: Logistic Regression Confusion Matrix

In general the ensemble based models achieved much better performance than the linear classifiers.

HistGradientBoosting

- Obtained high classification accuracy and balanced Recall.
- Efficiently executed TF-IDF genomic feature representations.

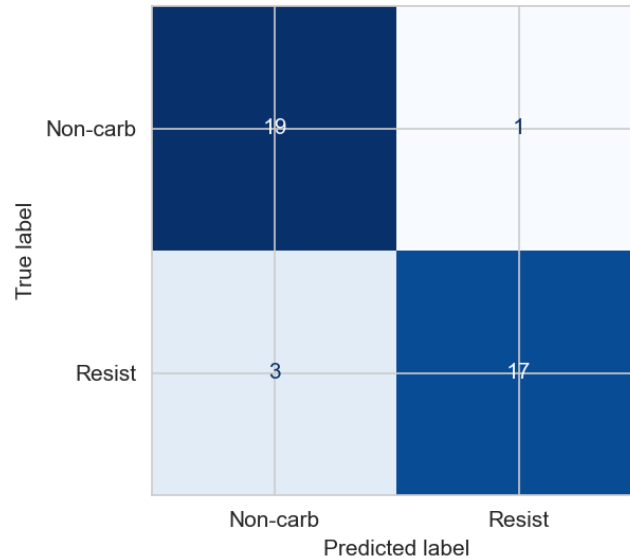


Figure 9: HistGradientBoosting Confusion Matrix

LinearSVC

- It is able to accurately categorize high-dimensional TF-IDF DNA features.
- Slightly less effective than ensemble-based classifiers, which make simpler linear separation assumptions.

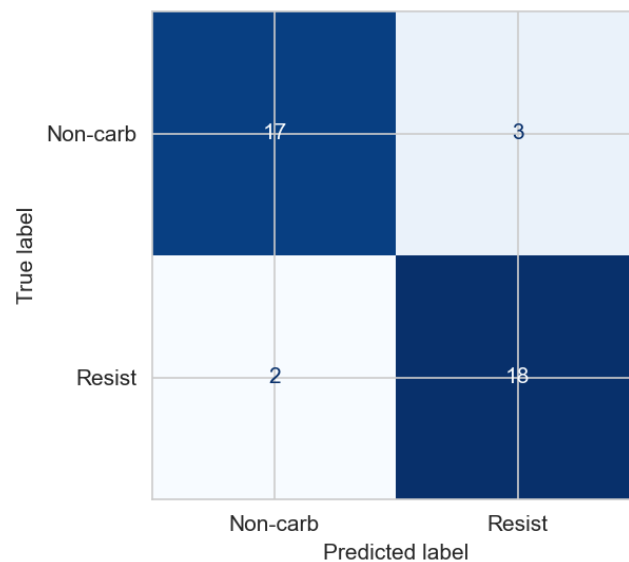


Figure 10: LinearSVC Confusion Matrix

4.7 Discussion

The results showed that machine learning algorithms designed to predict carbapenem resistance is possible to achieve using DNA sequence.

Key observations:

- Unrealistic 100% accuracy was avoided by strict homology filtering Motifs of the genome were captured well at the character level.
- Genomic motifs were captured well at the character level with 4-mers.
- Ensemble models were found to be generalized better than linear models.
- Leakage auditing enhanced the evaluation with increased reliability

This pipeline focused on getting realistic results and not artificially increase accuracy.

5. Conclusion

The objective of this project was to create a machine learning model that is capable of predicting the resistance to carbapenems in *Klebsiella pneumoniae* from genomic DNA sequences that is robust and leakage controlled.

The pipeline integrated:

- NCBI sequence retrieval
- Data cleaning
- Homology reduction
- TF-IDF-based 4-mer extraction
- SMOTE balancing
- Multi-model classification

The best performance was obtained by Random Forest with:

- Accuracy = 95%
- F1 Score = 95%
- ROC-AUC = 99.5%

The project points out the significance of leakage prevention and realistic appraisal of machine learning systems in the field of bioinformatics.

Future improvements may include:

- DNABERT transformers
- Deep learning architectures
- Clinical MIC validation datasets
- Multi-antibiotic resistance prediction
- SHAP-based explainability

This is a proof-of-concept for the use of computational genomics to perform fast screening of AMR and precision medicine applications.

6. References

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